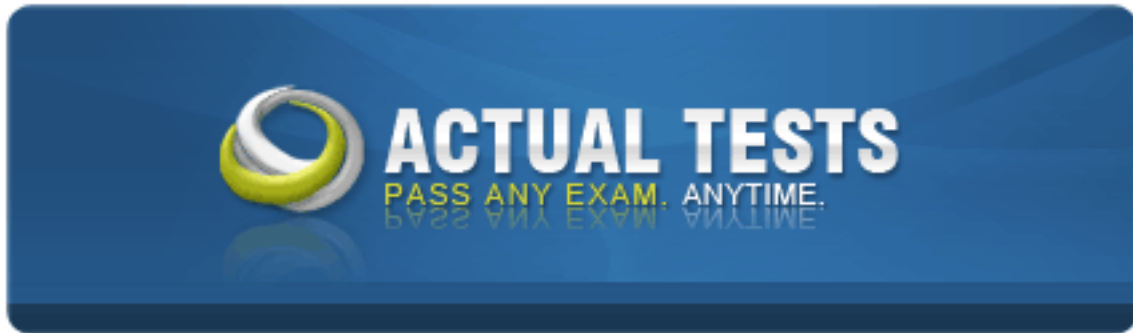


Cisco 640-875



**Building Cisco Service Provider Next-Generation
Networks, Part 1
Version: 6.0**

QUESTION NO: 1

Given the IP address 172.16.170.15 255.255.224.0, which three options are the subnet that this IP address belongs to, the host IP address range on this subnet, and the subnet broadcast IP address? (Choose three.)

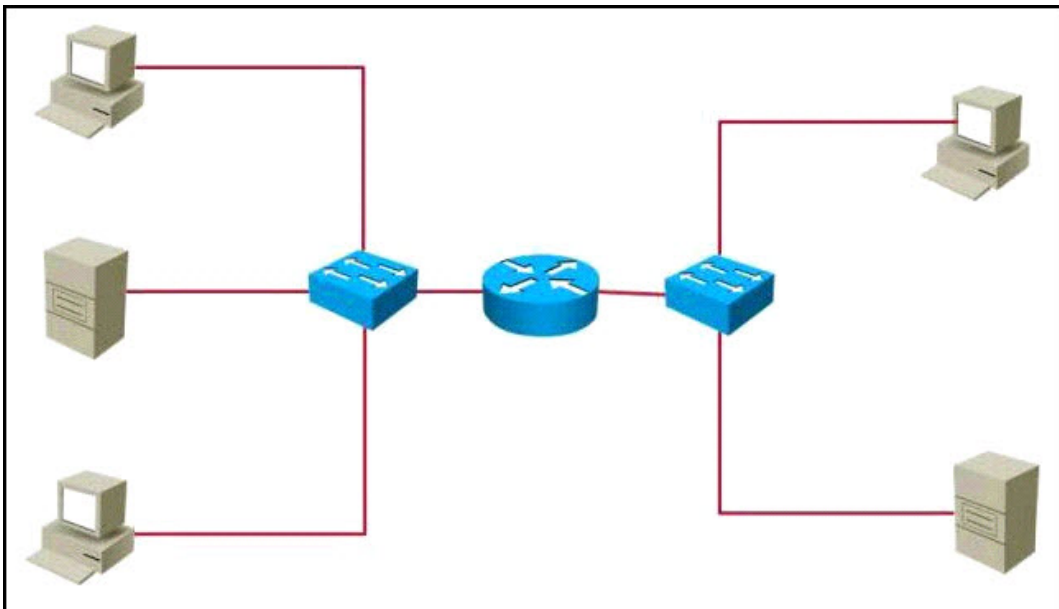
- A. subnet = 172.16.160.0/19
- B. subnet = 172.16.170.0/19
- C. subnet = 172.16.192.0/19
- D. host IP address range = 172.16.160.1 to 172.16.191.254
- E. host IP address range = 172.16.170.1 to 172.16.170.254
- F. host IP address range = 172.16.160.1 to 172.16.223.254
- G. subnet broadcast IP address = 172.16.170.255
- H. subnet broadcast IP address = 172.16.191.255
- I. subnet broadcast IP address = 172.16.223.255

Answer: A,D,H

Explanation:

QUESTION NO: 2

Refer to the exhibit.



Which two options are the number of broadcast domains and collision domains shown in the exhibit? (Choose two.)

- A. 1 collision domain

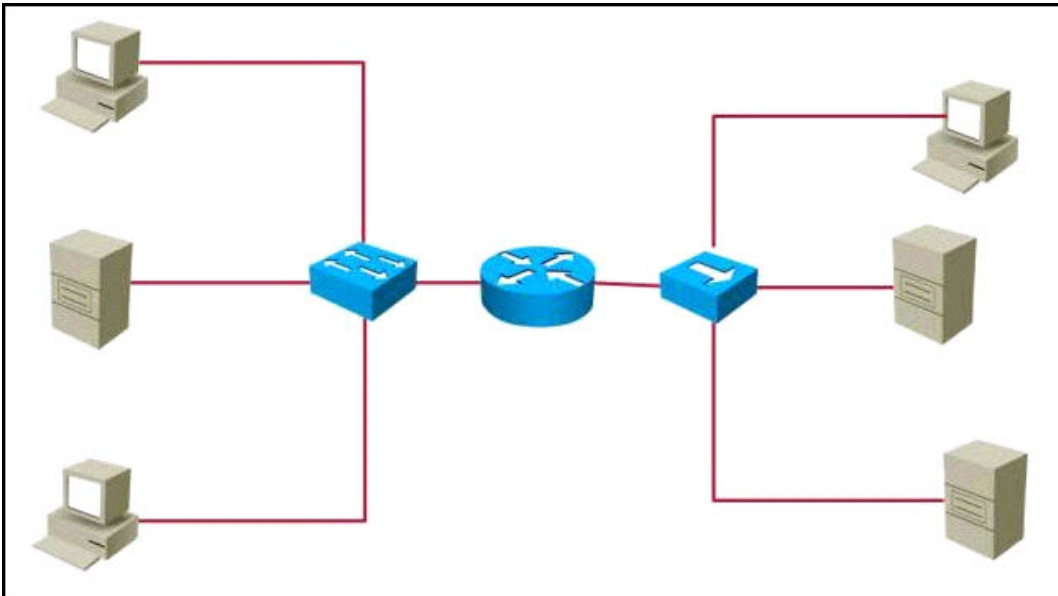
- B. 2 collision domains
- C. 4 collision domains
- D. 7 collision domains
- E. 1 broadcast domain
- F. 2 broadcast domains
- G. 4 broadcast domains
- H. 7 broadcast domains

Answer: D,F

Explanation:

QUESTION NO: 3

Refer to the exhibit.



Which two options are the number of broadcast domains and collision domains shown in the exhibit? (Choose two.)

- A. 2 collision domains
- B. 4 collision domains
- C. 5 collision domains
- D. 8 collision domains
- E. 2 broadcast domains
- F. 4 broadcast domains
- G. 5 broadcast domains
- H. 8 broadcast domains

Answer: C,E

Explanation:

QUESTION NO: 4

Which three IP addresses are valid for hosts that belong to the 10.152.0.0/13 subnet?

(Choose three.)

- A. 10.152.0.0/13
- B. 10.153.0.1/13
- C. 10.158.255.255/13
- D. 10.159.255.254/13
- E. 10.160.0.1/13
- F. 10.161.0.1/13

Answer: B,C,D

Explanation:

QUESTION NO: 5

Which type of IPv6 address does the ff00::/8 prefix denote?

- A. local loopback
- B. global unicast
- C. unique link local
- D. link local unicast
- E. multicast
- F. anycast

Answer: E

Explanation:

QUESTION NO: 6

With IPv6, for which purpose are router solicitation and router advertisement used?

- A. routing protocol updates

- B. routing protocol neighbor peerings
- C. router and prefix discovery
- D. Layer 3 to Layer 2 address resolution (similar to IPv4 ARP)

Answer: C

Explanation:

QUESTION NO: 7

Which statement about IPv6 global unicast addresses is true?

- A. The first 3 bits is 001.
- B. The first 4 bits is 0011.
- C. The first 4 bits is 1111.
- D. The first 16 bits is FE80 in hex.
- E. The first 16 bits is FF00 in hex.
- F. The first 16 bits is 2002 in hex.
- G. The first 16 bits is 3FFE in hex.

Answer: A

Explanation:

QUESTION NO: 8

Given the IP address 10.106.170.145 255.248.0.0, which three options are the subnet that this IP address belongs to, the host IP address range on this subnet, and the subnet broadcast IP address?

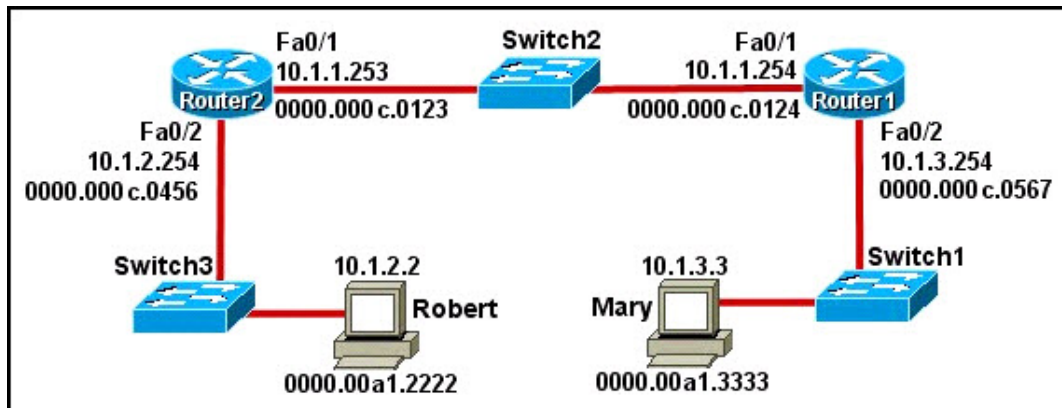
- A. subnet = 10.104.0.0/13
- B. subnet = 10.106.0.0/13
- C. subnet = 10.112.0.0/13
- D. host IP address range = 10.104.0.1 to 10.111.255.254
- E. host IP address range = 10.106.0.1 to 10.106.255.254
- F. host IP address range = 10.106.0.1 to 10.119.255.254
- G. subnet broadcast IP address = 10.106.255.255
- H. subnet broadcast IP address = 10.111.255.255
- I. subnet broadcast IP address = 10.119.255.255

Answer: A,D,H

Explanation:

QUESTION NO: 9

Refer to the exhibit.



Robert is sending an instant message to Mary. The message will be broken into a series of packets that will traverse all network devices. Which four options are the MAC and IP source and destination addresses that will populate these packets as they are forwarded from Router2 to Router1? (Choose four.)

- A. source MAC = 0000.00a1.2222
- B. source MAC = 0000.000c.0123
- C. destination MAC = 0000.00a1.3333
- D. destination MAC = 0000.000c.0124
- E. source IP 10.1.2.2
- F. source IP 10.1.1.253
- G. destination IP = 10.1.3.3
- H. destination IP = 10.1.1.254

Answer: B,D,E,G

Explanation:

QUESTION NO: 10

Which Layer 2 WAN protocol uses DLCI as the virtual circuit identifier?

- A. ATM
- B. PPP
- C. Frame Relay
- D. Packet over SONET

E. Metro Ethernet

Answer: C

Explanation:

QUESTION NO: 11

Which two statements about Packet over SONET are true? (Choose two.)

- A. POS is a Layer 2 technology that uses PPP in ATM encapsulation and SONET framing.
- B. The Layer 2 frame is encapsulated into a generic HDLC header (not Cisco proprietary HDLC) and placed into the appropriate SONET Payload Envelope.
- C. The FCS field can be set to 16 bits on one end and 32 bits on the other end of a POS connection without causing any connectivity issues.
- D. The Cisco POS interface supports Cisco HDLC and PPP encapsulation.
- E. POS can be used in a point-to-point or point-to-multipoint environment.

Answer: B,D

Explanation:

QUESTION NO: 12

Which pair of mechanisms are often used together to allow dynamic routing over a secured site-to-site VPN tunnel?

- A. GRE + SSL
- B. VTI + SSL
- C. VRF + SSL
- D. GRE + IPsec
- E. L2TP + IPsec
- F. VRF + IPsec

Answer: D

Explanation:

QUESTION NO: 13

Which two statements about IPv4 DHCP operations are true? (Choose two.)

- A. If the DHCP clients and servers are on the same subnet, DHCP uses UDP broadcasts to communicate between the clients and servers.
- B. The DHCP server that receives a DHCPDISCOVER message may respond with a DHCPOFFER message on UDP port 67.
- C. A client can receive DHCPOFFER from multiple servers and it can accept all the offers by broadcasting the DHCPREQUEST message to all the servers.
- D. A DHCP relay is required if the DHCP clients and the servers are on the same subnet.
- E. If the client and server are on different subnets, DHCPDISCOVER and DHCPREQUEST messages are sent via UDP broadcasts, but DHCPOFFER and DHCPACK messages are unicast.
- F. The client broadcasts a DHCPDISCOVER message on UDP port 68 with its own MAC address to discover available DHCP servers.

Answer: A,E

Explanation:

QUESTION NO: 14

Which Cisco router feature allows multiple independent routing table instances to co-exist within the same router at the same time?

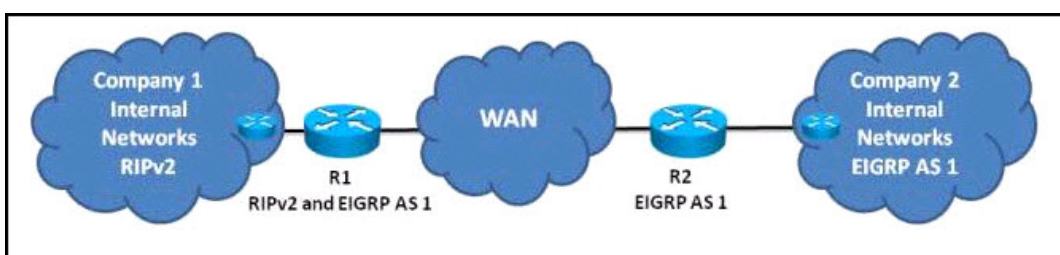
- A. Virtual Contexts
- B. Virtual Machines
- C. VTI
- D. VRF
- E. VPLS
- F. GRE

Answer: D

Explanation:

QUESTION NO: 15

Refer to the exhibit. Company 1 has merged with Company 2. Company 1 is using RIPv2 as its IGP and Company 2 is using EIGRP as its IGP. EIGRP also is running between R1 and R2 over the WAN link.



Which action must be taken to allow these two configurations?

- The Company 1 RIP routes appear as external EIGRP routes within Company 2.
 - The Company 2 EIGRP routes appear as RIPv2 routes within Company 1.
- A. Enable RIPv2 on the R2 router.
B. Enable RIPv2 on the R2 router and on all internal routers of Company 2. Enable EIGRP on all internal routers of Company 1.
C. Configure mutual route redistribution between RIPv2 and EIGRP on the R1 and R2 routers.
D. Configure mutual route redistribution between RIPv2 and EIGRP on the R1 router.
E. Configure mutual route redistribution between RIPv2 and EIGRP on the R2 router.

Answer: D

Explanation:

QUESTION NO: 16

Refer to the exhibit.

```
interface e0
ip address 192.168.1.1 255.255.255.0
ip nat outside
!
interface e1
ip address 10.1.1.1 255.255.255.0
ip nat inside
!
ip nat inside source static 10.1.1.2 192.168.1.2
```

Which two statements about these partial Cisco IOS/IOS-XE configurations are true? (Choose two.)

- A. 10.1.1.2 is the inside local address.
B. 192.168.1.2 is the outside local address.
C. 192.168.1.2 is the outside global address.
D. This is an example of static one-to-one Network Address Translation.
E. This is an example of static Port Address Translation.

Answer: A,D

Explanation:

QUESTION NO: 17

Refer to the exhibit.

RouterX# **show ip nat translations**

Pro	Inside global	Inside local	Outside local	Outside global
TCP	172.17.38.1:1050	192.168.3.7:1050	10.1.1.1:23	10.1.1.1:23
TCP	172.17.38.1:1776	192.168.4.12:1776	10.2.2.2:25	10.2.2.2:25

Which configuration is necessary to support this output?

A. interface Ethernet0

B. ip address 192.168.3.1 255.255.255.0

ip nat outside

!

interface Ethernet1

ip address 192.168.4.1 255.255.255.0

ip nat outside

!

interface Serial0

ip address 172.17.38.1 255.255.255.0

ip nat inside

!

ip nat inside source list 1 interface Serial0 overload

!

access-list 1 permit 192.168.3.0 0.0.0.255

access-list 1 permit 192.168.4.0 0.0.0.255

interface Ethernet0

C. ip address 192.168.3.1 255.255.255.0

ip nat inside

!

interface Ethernet1

ip address 192.168.4.1 255.255.255.0

ip nat inside

!

interface Serial0

ip address 172.17.38.1 255.255.255.0

```
ip nat outside
!  
ip nat inside source list 1 interface Serial0 overload
!  
access-list 1 permit 172.17.38.0 0.0.0.255
```

```
interface Ethernet0
D. ip address 192.168.3.1 255.255.255.0
ip nat inside
!  
interface Ethernet1
ip address 192.168.4.1 255.255.255.0
ip nat inside
!  
interface Serial0
ip address 172.17.38.1 255.255.255.0
ip nat outside
!  
ip nat inside source list 1 static 172.17.38.1
!  
access-list 1 permit 192.168.3.0 0.0.0.255
access-list 1 permit 192.168.4.0 0.0.0.255
interface Ethernet0
E. ip address 192.168.3.1 255.255.255.0
ip nat inside
!  
interface Ethernet1
ip address 192.168.4.1 255.255.255.0
ip nat inside
!  
interface Serial0
ip address 172.17.38.1 255.255.255.0
ip nat outside
!  
ip nat inside source list 1 interface Serial0 overload
!  
access-list 1 permit 192.168.3.0 0.0.0.255
access-list 1 permit 192.168.4.0 0.0.0.255
```

Answer: D

Explanation:

QUESTION NO: 18

Which IPv6 feature replaced and enhanced the IPv4 ARP functions?

- A. ARPv6
- B. DHCPv6
- C. neighbor discovery protocol in ICMPv6
- D. router solicitation and router advertisement
- E. autoconfiguration

Answer: C

Explanation:

QUESTION NO: 19

Which three statements correctly describe DSL-based Internet services for residential users? (Choose three.)

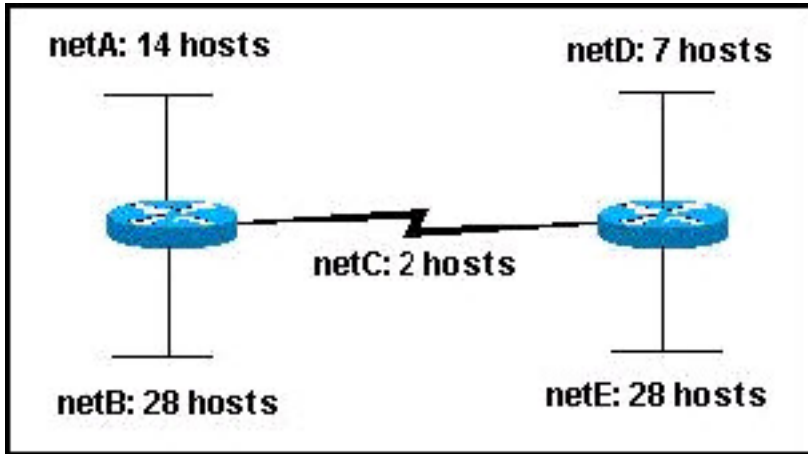
- A. Most residential deployments use PPPoE encapsulation where the end user PC or the DSL CPE will require a PPP username and password configuration.
- B. Most residential deployments use PPPoA encapsulation where the DSL CPE device needs to provide Layer 3 routing functions.
- C. The most commonly installed DSL variant for residential deployments is ADSL.
- D. The most commonly installed DSL variant for residential deployments is SDSL.
- E. The most commonly installed DSL variant for residential deployments is HDSL.
- F. At the DSL service provider location, a DSLAM typically is used to terminate all the PPP connections of the DSL subscribers.
- G. At the DSL service provider location, a DSLAM is used to terminate and then aggregate the DSL connections over an ATM network.

Answer: A,C,G

Explanation:

QUESTION NO: 20

Refer to the exhibit.



Given the Class C IP network of 192.168.1.0, using VLSM, which five subnets meet the requirements in the exhibit? (Choose five.)

- A. 192.168.1.0/27
- B. 192.168.1.32/27
- C. 192.168.1.48/27
- D. 192.168.1.48/28
- E. 192.168.1.64/28
- F. 192.168.1.80/28
- G. 192.168.1.80/29
- H. 192.168.1.88/30
- I. 192.168.1.96/30

Answer: A,B,E,F,I

Explanation:

QUESTION NO: 21

Refer to the exhibit.

Host A ---- Router A ----- Router B ----- Host B

Which layer(s) of the TCP/IP model operates end-to-end (Host A to Host B)?

- A. physical layer
- B. link layer
- C. transport and application layers
- D. link and internet (network) layers

E. physical and link layers

Answer: C

Explanation:

QUESTION NO: 22

Which two applications require the network to have the least amount of jitter, delay, and packet loss? (Choose two.)

- A. FTP
- B. SSH
- C. VoIP
- D. email
- E. Cisco TelePresence
- F. web browsing

Answer: C,E

Explanation:

QUESTION NO: 23

What is the default spanning-tree mode of an NNI port on a Cisco ME3400 switch?

- A. multiple spanning tree (mst)
- B. per-vlan rapid spanning tree (rapid-pvst)
- C. per-vlan spanning tree (pvst)
- D. IEEE 802.1D (stp)

Answer: B

Explanation:

QUESTION NO: 24

Which two statements about transparent bridging operations are true? (Choose two.)

- A. A bridge builds its MAC address table dynamically using the spanning-tree protocol.
- B. A bridge operates at the data link layer.

- C. A bridge sends unknown unicast frames out on all ports except the incoming port.
- D. A bridge forwards traffic based on the destination IP address.
- E. All the ports on the bridge belong to the same collision domain.

Answer: B,C

Explanation:

QUESTION NO: 25

Refer to the exhibit.

```
SW1#show port-security interface fa0/1

Port Security: Enabled
Port Status: Secure-up
Violation Mode: Shutdown
Aging Time: 0 mins
Aging Type: Absolute
SecureStatic Address Aging: Disabled
Maximum MAC Addresses: 10
Total MAC Addresses: 1
Configured MAC Addresses: 0
Sticky MAC Addresses: 0
Last Source Address:Vlan: e8b7.482c.a180:1
Security Violation Count: 0
```

According to this show output, what are sticky MAC addresses?

- A. dynamically learned MAC addresses on the port that are stored in the MAC address table and also in the start-up configuration
- B. statically configured MAC addresses on the port that are stored in the MAC address table
- C. statically configured MAC addresses on the port that are stored in the start-up configuration
- D. dynamically learned MAC addresses on the port that are stored in the running configuration

Answer: D

Explanation:

QUESTION NO: 26

Which two statements about classful or classless routing protocol operations are true? (Choose two.)

- A. Classful routing protocols do not send subnet mask information with their routing updates.
- B. Variable-length subnet masks are supported when using classful routing protocols.
- C. Examples of classless routing protocols include RIPv1, EIGRP, OSPF, and IS-IS.
- D. Classful routing protocols do not support discontinuous subnets.

Answer: A,D

Explanation:

QUESTION NO: 27

When two or more different routes exist to the same destination that have been learned from two different routing protocols, what does a router use to determine the best route?

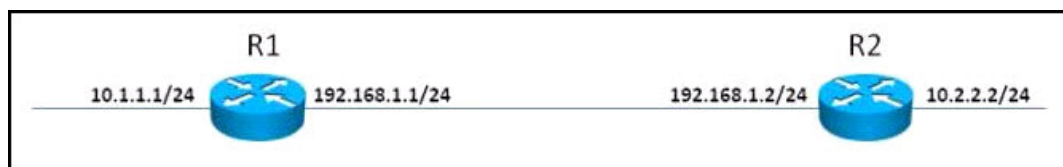
- A. routing protocol metric
- B. administrative distance
- C. hop count
- D. path cost
- E. local preference

Answer: B

Explanation:

QUESTION NO: 28

Refer to the exhibit.



R1 and R2 are running RIPv1 for the 10.0.0.0 and 192.168.1.0 networks. No default or static routes have been configured on the routers. Which two statements about this topology are true? (Choose two.)

- A.** R1 summarizes the 10.1.1.0/24 subnet to 10.0.0.0/8 when it sends its RIPv1 routing updates to R2. R2 summarizes the 10.2.2.0/24 subnet to 10.0.0.0/8 when it sends its RIPv1 routing updates to R1.
- B.** To support this topology, R1 and R2 must be configured so that RIPv1 auto-summary is disabled.
- C.** R1 sends information about the 10.1.1.0/24 subnet when it sends its RIPv1 routing updates to R2. R2 sends information about the 10.2.2.0/24 subnet when it sends its RIPv1 routing updates to R1. Example of RIPv1 update from R1 to R2:
- D.** RIP: build update entries
network 10.1.1.0/24 metric 1
- Hosts on the 10.1.1.0/24 subnet are unable to communicate with hosts on the 10.2.2.0/24 subnet.
- E.** R1 accepts the 10.2.2.0/24 RIPv1 routing updates from R2 and installs the 10.2.2.0/24 route in its routing table. R2 accepts the 10.1.1.0/24 RIPv1 routing updates from R1 and installs the 10.1.1.0/24 route in its routing table.

Answer: A,D

Explanation:

QUESTION NO: 29

Which Cisco IOS XR CLI command displays the running state of each line card (including the fan tray) on the Cisco ASR 9000 Series Router?

- A.** RP/0/RSP0/CPU0:PE1#show platform
- B.** RP/0/RSP0/CPU0:PE1(admin)#show platform
- C.** RP/0/RSP0/CPU0:PE1#show hw-module fpd location all
- D.** RP/0/RSP0/CPU0:PE1(admin)#show hw-module fpd location all
- E.** RP/0/RSP0/CPU0:PE1#show version
- F.** RP/0/RSP0/CPU0:PE1(admin)#show version

Answer: B

Explanation:

QUESTION NO: 30

Which Cisco IOS XR CLI command can you use to determine if the firmware on any of the line cards on the Cisco ASR 9000 Series Router needs to be upgraded?

- A.** RP/0/RSP0/CPU0:PE1#show platform
- B.** RP/0/RSP0/CPU0:PE1(admin)#show platform
- C.** RP/0/RSP0/CPU0:PE1#show hw-module fpd location all

- D. RP/0/RSP0/CPU0:PE1#show version
- E. RP/0/RSP0/CPU0:PE1(admin)#show version
- F. RP/0/RSP0/CPU0:PE1#show install active

Answer: C

Explanation:

QUESTION NO: 31

Which Cisco IOS CLI command is used to verify the speed and duplex settings on the Fa0/1 port on a Cisco switch?

- A. show ip interfaces fa0/1
- B. show interfaces fa0/1
- C. show interfaces fa0/1 switchport
- D. show ip interfaces brief
- E. show interfaces switchport

Answer: B

Explanation:

QUESTION NO: 32

Refer to the exhibit.

```
S78#show monitor session 1

Session 1
-----
Type: Local Session
Source Ports:
  RX Only: Fa0/3
  TX Only: Fa0/4
  Both: Fa0/1-2
Destination Ports: Gi0/1
Encapsulation: Native
Ingress: Disabled
```

Which statement about this show command output is true?

- A. The S78 switch is performing remote-SPAN.
- B. All traffic that ports Fa0/1, Fa0/2, Fa0/3, and Fa0/4 send and receive is monitored.
- C. All traffic that port Gi0/1 sends and receives is monitored.
- D. The sniffer (network analyzer) should be connected to the Gi0/1 port.
- E. The SPAN destination interface replicates the source interface encapsulation method.

Answer: D

Explanation:

QUESTION NO: 33

Which three statements about Cisco Discovery Protocol operations on a Cisco ME3400 switch are true? (Choose three.)

- A. Cisco Discovery Protocol is enabled by default on NNI ports.
- B. Cisco Discovery Protocol is enabled by default on all NNI, ENI, and UNI ports.
- C. Cisco Discovery Protocol is disabled by default on ENI ports.
- D. Cisco Discovery Protocol is not supported on NNI or ENI ports.
- E. Cisco Discovery Protocol is not supported on UNI ports.

Answer: A,C,E

Explanation:

QUESTION NO: 34

Currently, a Cisco IOS XR router is configured to send IP SLA probes to a network management station that is acting as the IP SLA target device. How can IP SLA measurement accuracy be improved and also provide additional statistics?

- A. Use ICMP echo probes instead of UDP echo probes.
- B. Configure another Cisco router as the target and enable that other Cisco router for IP SLA Responder operations.
- C. Increase the rate (frequency) at which a specified IP SLA operation is sent into the network.
- D. Use UDP echo probes instead of ICMP echo probes and enable UDP echo service on the target network management station.
- E. Enable IP SLA Responder service on the target network management station

Answer: B

Explanation:

QUESTION NO: 35

Refer to the exhibit.

```
PE8#show logging
```

```
Syslog logging: enabled (0 messages dropped, 4 messages rate-limited, 0 flushes, 0 overruns, xml disabled, filtering disabled)
```

```
No Active Message Discriminator.
```

```
No Inactive Message Discriminator.
```

```
Console logging: level emergencies, 169 messages logged, xml disabled,  
filtering disabled
```

```
Monitor logging: level debugging, 0 messages logged, xml disabled,  
filtering disabled
```

```
Buffer logging: disabled, xml disabled,  
filtering disabled
```

```
Exception Logging: size (8192 bytes)
```

```
Count and timestamp logging messages: disabled
```

```
Persistent logging: disabled
```

```
No active filter modules.
```

```
Trap logging: level critical, 127 message lines logged
```

```
Logging to 1.1.1.1 (udp port 514, audit disabled,  
link up),  
2 message lines logged,  
0 message lines rate-limited,  
0 message lines dropped-by-MD,  
xml disabled, sequence number disabled  
filtering disabled
```

Which three statements about this show output are true? (Choose three.)

- A. Logging to the circular logging buffer has been disabled.
- B. All severity levels messages can be logged to the terminal lines other than the console .
- C. The logging facility has been set to "critical".
- D. Messages with Severity Levels 0, 1, and 2 are sent to the syslog server.
- E. 169 messages have been sent to the syslog server.
- F. The logging buffer size is set to 8192 bytes.

Answer: A,B,D

Explanation:

QUESTION NO: 36

Which two statements about NetFlow support on Cisco IOS XR are true? (Choose two.)

- A. Cisco IOS XR supports NetFlow version 9 export format only.
- B. NetFlow version 9 is a flexible and extensible means to carry NetFlow records from a network node to a collector.
- C. IP flow is based on a set of four parameters only (source IP address, destination IP address, source port, and destination port).
- D. A NetFlow flow is the bidirectional sequence of packets on a single interface.

Answer: A,B

Explanation:

QUESTION NO: 37

What are the four steps to enable SSH v1 and v2 server on Cisco IOS XR routers? (Choose four.)

- A. Configure a host name.
- B. Configure a domain name.
- C. Generate a crypto key pair.
- D. Import the router identity certificate and the CA root certificate to the router.
- E. Configure an ISAKMP policy.
- F. Bring up an SSH server using the ssh server command.

Answer: A,B,C,F

Explanation:

QUESTION NO: 38

Generally, which way is the fastest to open S3 and S4 service requests to the Cisco TAC?

- A. Use the Smart Call Home feature.
- B. Use the online TAC Service Request Tool.
- C. Call Cisco TAC on the phone.
- D. Email Cisco TAC with a high-priority email.
- E. Export the related Syslog/NetFlow information to TAC directly.

Answer: B

Explanation:

QUESTION NO: 39

Which two statements about Cisco IOS XR NTP configuration are true? (Choose two.)

- A. A network device can obtain NTP time information in three ways: polling the NTP server, listening to NTP broadcasts, and listening to NTP multicasts.
- B. NTP authentication supports MD5, SHA1, or SHA2.
- C. Client mode is a peer-to-peer mode.

- D. The two types of poll-based NTP associations are client mode and symmetric active mode.
- E. The Cisco IOS-XR implementation of NTP supports stratum 1 service.
- F. In client mode, clients can dynamically learn the NTP server IP address by listening to the NTP server broadcasts or multicasts.

Answer: A,D

Explanation:

QUESTION NO: 40

Which SNMPv2/SNMPv3 operation is not supported in Cisco IOS XR software?

- A. get-bulk
- B. get-request
- C. inform
- D. trap
- E. set-request

Answer: C

Explanation:

QUESTION NO: 41

Which three statements about the Cisco IOS XR task-based user authorization method are true? (Choose three.)

- A. A user's association with a particular user group links that user to a particular set of task IDs. A user that is associated with a task ID can execute any operation associated with that task ID.
- B. Users can be assigned only to one user group.
- C. Every user group is associated with one or more task groups.
- D. Every task group is defined by a set of task IDs.
- E. Each user is assigned a privilege level.
- F. Each IOS XR CLI command is assigned a privilege level.
- G. A task ID maps the IOS XR command to a privilege level.

Answer: A,C,D

Explanation:

QUESTION NO: 42

Which three statements about Cisco IOS XR CLI operations are true? (Choose three.)

- A. Cisco IOS XR uses a three-stage configuration method.
- B. Committed changes are persistent across reboots in Cisco IOS XR.
- C. The commit replace global configuration command replaces the contents of the running configuration with the target configuration.
- D. The save global configuration command saves the contents of target configuration to the running configuration.
- E. There is no concept of a startup-config in Cisco IOS XR.
- F. The clear global configuration command erases the entire current running configuration.
- G. The show running-config global configuration command displays the configuration that occurs if the contents of the uncommitted changed (target configuration) are committed to the running configuration.

Answer: B,C,E

Explanation:

QUESTION NO: 43

What is the result of using the Cisco IOS copy tftp: run command?

- A. The running configuration on the Cisco IOS device is backed up to the TFTP server.
- B. Only new configuration parameters from the saved configuration file on the TFTP server are added to the running configuration. No changes to any existing configuration parameters in the running configuration are made.
- C. The running configuration on the Cisco IOS device is replaced with the saved configuration file on the TFTP server.
- D. The configuration stored in the TFTP server is merged with the running configuration on the Cisco IOS device.
- E. The saved configuration file from the TFTP server is copied to the target configuration.

Answer: D

Explanation:

QUESTION NO: 44

Which component is integrated into the Cisco IPoDWDM interfaces?

- A. ROADM

- B. transponder
- C. electrical cross connect
- D. E-O-E converter
- E. SONET/SDH controller

Answer: B

Explanation:

QUESTION NO: 45

What is the SDH equivalent of the SONET STS-12 framing format?

- A. SDH STM-1
- B. SDH STM-4
- C. SDH STM-12
- D. SDH STM-16
- E. SDH STM-64

Answer: B

Explanation:

QUESTION NO: 46

What are the three RFC 1918 IPv4 private address ranges? (Choose three.)

- A. 10.0.0.0 - 10.255.255.255
- B. 172.16.0.0 - 172.255.255.255
- C. 172.16.0.0 - 172.31.255.255
- D. 192.168.0.0 - 192.168.31.255
- E. 192.168.0.0 - 192.168.255.255

Answer: A,C,E

Explanation:

QUESTION NO: 47

With a distance vector routing protocol like RIPv1, when is a route in the routing table placed in the "possibly down" state?

- A. when a router receives an update from its neighbor router about a network in its routing table but with a worse metric
- B. when a router receives an update from its neighbor router about a network in its routing table but with a better metric
- C. when a router receives an update from its neighbor router indicating that a previously accessible network is now inaccessible
- D. whenever the router receives an update from its neighbor about a newly learned network
- E. whenever the router interface that is used to reach the remote network goes down

Answer: C

Explanation:

QUESTION NO: 48

When EIGRP is enabled on Cisco IOS XR, which EIGRP feature is on by default?

- A. no auto-summary
- B. stub routing
- C. unequal cost paths load-balancing
- D. no ip eigrp split-horizon
- E. eigrp authentication

Answer: A

Explanation:

QUESTION NO: 49

Which feature is used to protect the control plane of the router?

- A. port security
- B. uRPF
- C. routing protocol authentication
- D. VRF
- E. SSH
- F. SSL

Answer: C

Explanation:

QUESTION NO: 50

Given the IP address of 209.168.200.155/27, which two options are the subnet that this IP address belongs to and the subnet broadcast IP address? (Choose two.)

- A. 209.168.200.64 255.255.255.224
- B. 209.168.200.96 255.255.255.224
- C. 209.168.200.128 255.255.255.224
- D. 209.168.200.95
- E. 209.168.200.127
- F. 209.168.200.159

Answer: C,F

Explanation:

QUESTION NO: 51

Given the CIDR address range of 200.165.160.0 to 200.165.191.255, what is the CIDR block?

- A. 200.165.160.0/17
- B. 200.165.160.0/18
- C. 200.165.160.0/19
- D. 200.165.160.0/20
- E. 200.165.160.0/21

Answer: C

Explanation:

QUESTION NO: 52

Given these four Class B networks: 172.104.0.0, 172.105.0.0, 172.106.0.0, 172.107.0.0, what is the supernet (CIDR) address that summarizes these four Class B networks?

- A. 172.104.0.0/12
- B. 172.104.0.0/13
- C. 172.104.0.0/14
- D. 172.104.0.0/15

Answer: C

Explanation:

QUESTION NO: 53

This show output is from the S7 Cisco Metro Switch and the PE7 Cisco ASR 9000 Series Router. Which configuration change(s) is/are needed to bring the S7 FastEthernet 0/2 interface into the up/up state?

```
S7#show cdp neighbors
<OUTPUT OMITTED>
```

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
PE7	Fas 0/2	163	R	ASR9K Ser	Gig 0/0/0/0
S78	Fas 0/23	146	S I	ME-3400E-	Fas 0/21
S8	Fas 0/21	178	S I	ME-3400E-	Fas 0/21

*Mar 20 21:28:02.177: %ETHCNTR-3-HALF_DUX_COLLISION_EXCEED_THRESHOLD.
Collisions at FastEthernet0/2 exceed threshold. Considered as loop-back.

*Mar 20 21:28:02.177: %PM-4-ERR_DISABLEF. loopback error detected on Fa0/2, putting Fa0/2 in
err-disable state

*Mar 20 21:28:03.184: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to down

```
S7#show interface fa0/2
```

FastEthernet0/2 is down, line protocol is down (err-disabled)

Hardware is Fast Ethernet, address is e8ba.70b5.6c04 (bia e8ba.70b5.6c04)

MTU 1500 bytes, BW 10000 Kbit, DLY 1000 usec,

reliability 255/255, txload 1/255, rxload 1/255

Encapsulation ARPA, loopback not set

Keepalive set (10 sec)

Half-duplex, 10Mb/s, media type is 10/100BaseTX

input flow-control is off, output flow-control is unsupported

ARP typF. ARPA, ARP Timeout 04:00:00

Last input 00:00:40, output 00:00:00, output hang never

Last clearing of "show interface" counters never

Input queueF. 0/75/0/0 (size/max/drops/flushes); Total output drops: 0

Queueing strategy: fifo

Output queueF. 0/40 (size/max)

5 minute input rate 0 bits/sec, 0 packets/sec

5 minute output rate 0 bits/sec, 0 packets/sec

484423 packets input, 39349929 bytes, 0 no buffer

Received 484423 broadcasts (167203 multicasts)

0 runts, 0 giants, 0 throttles

1 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored

0 watchdog, 167203 multicast, 0 pause input

0 input packets with dribble condition detected

972398 packets output, 128931949 bytes, 0 underruns

0 output errors, 46 collisions, 4 interface resets

0 babbles, 0 late collision, 0 deferred

0 lost carrier, 0 no carrier, 0 PAUSE output

0 output buffer failures, 0 output buffers swapped out

RP/0/RSP0/CPU0:PE7#show interfaces gi0/0/0/0

GigabitEthernet0/0/0/0 is down, line protocol is down

Interface state transitions: 4

Hardware is GigabitEthernet, address is 4055.392f.40a8 (bia 4055.392f.40a8)

Internet address is 192.168.107.70/24

MTU 1514 bytes, BW 100000 Kbit (Max: 100000 Kbit)

reliability 255/255, txload 0/255, rxload 0/255

Encapsulation ARPA,

Full-duplex, 100Mb/s, THD, link type is force-up

output flow control is off, input flow control is off

loopback not set,

ARP type ARPA, ARP timeout 04:00:00

Last input 00:00:00, output 00:00:35

Last clearing of "show interface" counters never

5 minute input rate 0 bits/sec, 0 packets/sec

5 minute output rate 0 bits/sec, 0 packets/sec

972551 packets input, 128953765 bytes, 729 total input drops

172486 drops for unrecognized upper-level protocol

Received 0 broadcast packets, 800781 multicast packets

0 runts, 0 giants, 0 throttles, 0 parity

0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort

484545 packets output, 39360900 bytes, 0 total output drops

Output 317283 broadcast packets, 167262 multicast packets

0 output errors, 0 underruns, 0 applique, 0 resets

0 output buffer failures, 0 output buffers swapped out

4 carrier transitions

- A. Enable loopback line on the S7 Fa0/2 interface then shut and no shut the Fa0/2 interface.
- B. Enable duplex half on the PE7 Gi0/0/0/0 interface then shut and no shut the Fa0/2 interface on S7.
- C. Change S7 Fa0/2 interface to duplex full and speed 100 then shut and no shut the Fa0/2 interface.
- D. Just need to no shut the Fa0/2 interface on S7.
- E. Just need to shut and no shut the Fa0/2 interface on S7.

Answer: C

Explanation:

QUESTION NO: 54

What does a transparent LAN switch use to build its CAM table?

- A. BPDU messages exchanged with neighbor bridges
- B. STP messages exchanged with neighbor bridges
- C. dynamic routing protocol updates from neighbor bridges
- D. analysis of the source MAC address of incoming frames on the bridge ports

Answer: D

Explanation:

QUESTION NO: 55

You are having problems browsing to <http://www.cisco.com> from your laptop that is running Windows. Your laptop is connected directly to a default gateway, which is a Cisco ISR G2 router. Which four options are the basic troubleshooting steps that you should use to troubleshoot this issue? (Choose four.)

- A. Issue the ipconfig command from the Windows command line on your laptop to determine your laptop IP address and the default gateway IP address.
- B. Issue the ipconfig /all command from the Windows command line on your laptop to determine the DNS server IP address.
- C. Issue the ping {default gateway IP address} command from the Windows command line on your laptop to verify connectivity to the default gateway.
- D. Issue the ping {DNS server IP address} command from the Windows command line on your laptop to verify connectivity to the DNS server.
- E. Issue the ping tcp www.cisco.com 80 command from the Windows command line on your laptop to verify TCP port 80 connectivity to the www.cisco.com server.
- F. From the ISR G2 CLI, issue the ping tcp www.cisco.com 80 command to verify TCP port 80 connectivity to the www.cisco.com server.
- G. From the ISR G2 CLI, issue the ISR-G2#tracert www.cisco.com command to verify connectivity to the www.cisco.com server.
- H. From the ISR G2 CLI, issue the ISR-G2#nslookup www.cisco.com command to determine the IP address of the www.cisco.com server.

Answer: A,B,C,D

Explanation:

QUESTION NO: 56

Refer to the exhibit.

This is a partial router configuration:

```
!partial router config
!  
hostname R1  
!  
interface fa0/1  
no shut  
!  
interface fa0/2  
ip address 192.168.101.1 255.255.255.0  
no shut  
!
```

You want to enable these three actions:

- Enable the fa0/1 router's interface to obtain its interface IP address dynamically from a DHCP server.
- Enable the router to configure the next hop ip address of the default route automatically based on the DHCP response.
- Enable the router to act as the DHCP server for other clients connected to the router fa0/2 interface.

Which three Cisco IOS configurations must you use? (Choose three.)

- A.** ip route 0.0.0.0 0.0.0.0 fa0/1
- B.** ip route 0.0.0.0 0.0.0.0 dhcp
- C.** ip dhcp excluded-address 192.168.101.1
- D.** !
ip dhcp pool test1
network 192.168.101.0 255.255.255.0
default-router 192.168.101.1
dns-server 200.165.200.101
ip dhcp pool test2
- E.** ip local pool test2 192.168.101.2 192.168.101.254
default-router 192.168.101.1
dns-server 200.1665.200.101
int fa0/1
- F.** ip address auto
int fa0/1

- G. ip address dhcp
int fa0/1
- H. ip helper-address dhcp

Answer: B,C,F

Explanation:

QUESTION NO: 57

You need to enable the "local-router" serial 0/0 connection to the "remote-router" to be authenticated using the "C!sco!23" password. The password cannot be sent in cleartext between the two routers. The "remote-router" has already been configured with all the required configurations.

Which four Cisco IOS configurations are necessary? (Choose four.)

- A. hostname local-router
- B. int s0/0
- C. encap hdlc
int s0/0
- D. encap ppp
int s0/0
- E. ppp authentication pap
int s0/0
- F. ppp authentication chap
ip domain-name test.com
- G. username remote-router password C!sco!23
- H. username local-router password C!sco!23

Answer: A,C,E,G

Explanation:

QUESTION NO: 58

Which two statements about Cisco IOS XE operations are true? (Choose two.)

- A. Cisco IOS XE uses a two-stage configuration method.
- B. Using the commit global configuration command, all the committed changes will be persistent across reboots.

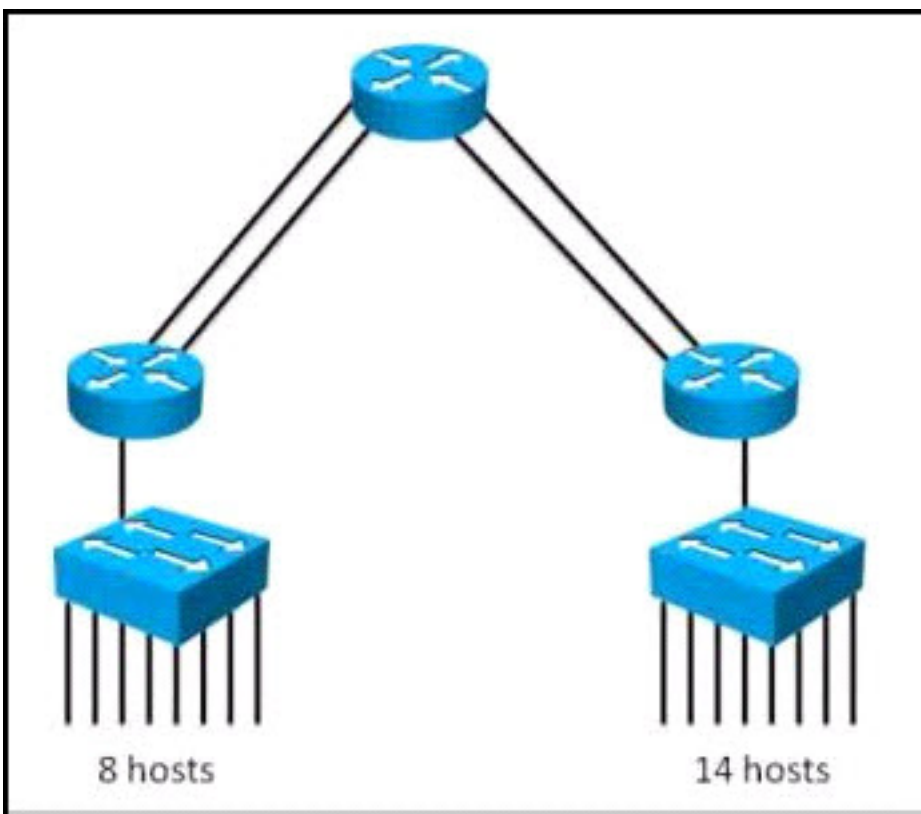
- C. The copy run start command makes the configuration changes persistent across reboots.
- D. The erase start command followed by the reboot command brings up the router without any configurations.
- E. There is no concept of a startup-config in Cisco IOS XE.

Answer: C,D

Explanation:

QUESTION NO: 59

Refer to the exhibit.



Using a single Class C network of 192.168.1.0/24 and VLSM, which two options specify the number of subnets that are needed to support this network and what are the minimum sizes of the required subnets? (Choose two.)

- A. 4 subnets
- B. 6 subnets
- C. 8 subnets
- D. 26 subnets
- E. 28 subnets
- F. two /30 subnets + one /28 subnets + one /29 subnets
- G. four /30 subnets + two /28 subnets

- H. six /30 subnets + two /28 subnets
- I. twenty six /30 subnets
- J. twenty eight /30 subnets

Answer: B,G

Explanation:

QUESTION NO: 60

Which option is enabled by default on Cisco IOS and IOS XE routers?

- A. IPv6 unicast routing
- B. IPv4 unicast routing
- C. IPv6 and IPv4 unicast routing
- D. SSH server
- E. HTTP server
- F. SSH and HTTP server
- G. AAA console login authentication

Answer: B

Explanation:

QUESTION NO: 61

Using VLSM, how many /30 subnets can be derived from a single /27 Class C subnet?

- A. 1
- B. 2
- C. 4
- D. 8
- E. 12
- F. 16

Answer: D

Explanation:

QUESTION NO: 62

On Cisco IOS Switches, which port security violation mode results in frames from non-allowed

MAC address being dropped, log message created, and SNMP trap sent?

- A. protect
- B. restrict
- C. shutdown
- D. err-disabled
- E. log

Answer: B

Explanation:

QUESTION NO: 63

On Cisco IOS XR, for which purpose is the turboboot procedure used?

- A. to boot the router into ROM Monitor mode
- B. to install the IOS XR software from scratch using the ROM Monitor mode
- C. to install and boot up a PIE
- D. to install and boot up a SMU
- E. to perform an ISSU
- F. to speed up the time it takes to boot up the IOS XR router

Answer: B

Explanation:

QUESTION NO: 64

Which statement about 10/40/100 GigabitEthernet interface configuration on Cisco IOS XR is true?

- A. Use the negotiation auto command in interface configuration mode to enable link auto-negotiation to ensure that devices that share a link segment are automatically configured with the highest performance.
- B. The configuration supports only full-duplex operation.
- C. The range for MTU values is 64 bytes to 1518 bytes on the interface.
- D. The interface has no flow control configuration.

Answer: B

Explanation:

QUESTION NO: 65

Which three statements about the ARP function are true? (Choose three.)

- A. ARP is used to determine an IP address.
- B. An ARP request is broadcast at Layer 3.
- C. An ARP response is broadcast at Layer 2.
- D. ARP tables are maintained on Layer 3 devices.
- E. The destination MAC address of an ARP request is that of the ARP responder.
- F. ARP requests are broadcast at Layer 2.
- G. ARP resolves MAC addresses to NET addresses.
- H. ARP uses 0x0806 as the Type field in an Ethernet frame.

Answer: D,F,H

Explanation:

QUESTION NO: 66

If a bridge port is in the non-designated state, which two statements about spanning-tree operations on this port are true? (Choose two.)

- A. This port is in the Forwarding state.
- B. This port sends BPDUs on the attached link.
- C. This port has the lowest cost to the root bridge.
- D. This port is considered the root port.
- E. This port receives BPDUs.
- F. This port is in the Blocked state.

Answer: E,F

Explanation:

QUESTION NO: 67

Which statement about STP operations is true?

- A. One root bridge exists per collision domain.
- B. One root port is established per root bridge.
- C. One designated port is defined per segment.

- D. Non-designated ports are set in the forwarding state.
- E. One root port exists per broadcast domain.

Answer: C

Explanation:

QUESTION NO: 68

Which statement about a root bridge in an STP domain is true?

- A. It has the highest root ID in the STP domain.
- B. It has the lowest bridge ID in the STP domain.
- C. It contains non-designated ports.
- D. It contains root ports.
- E. It has the highest priority value.

Answer: B

Explanation:

QUESTION NO: 69

Which Cisco IOS XR command displays information about statically configured routes and their corresponding interface or next-hop?

- A. show ipv4 interfaces brief
- B. show route connected
- C. show interfaces Loopback
- D. show route static
- E. show route ospf

Answer: D

Explanation:

QUESTION NO: 70

Which statement about SSH is true?

- A. It is a TCP/IP transport layer protocol.

- B. It uses TCP port number 22.
- C. It securely encrypts Telnet.
- D. It uses UDP port 23.
- E. Passwords are sent in clear text.
- F. It is an OSI network layer protocol.

Answer: B

Explanation:

QUESTION NO: 71

Which Cisco IOS XR command displays traffic counters for an individual interface that is configured on the router?

- A. RP/0/RSP0/CPU0:PE1(config)# show bundle
- B. RP/0/RSP0/CPU0:PE1# show interfaces GigabitEthernet 0/0/0/1
- C. RP/0/RSP0/CPU0:PE1# show controllers GigabitEthernet 0/0/0/1
- D. RP/0/RSP0/CPU0:PE1(admin)# show platform
- E. RP/0/RSP0/CPU0:PE1(admin)# show hw-module fpd location all
- F. RP/0/RSP0/CPU0:PE1# show interfaces brief

Answer: B

Explanation:

QUESTION NO: 72

Which Cisco IOS XR command presents a list of all interfaces, their Layer 3 address (if configured), and the status of the interface and protocol?

- A. RP/0/RSP0/CPU0:PE1# show interface GigabitEthernet 0/0/0/1
- B. RP/0/RSP0/CPU0:PE1# show controllers
- C. RP/0/RSP0/CPU0:PE1(admin)# show hw-module fpd location all
- D. RP/0/RSP0/CPU0:PE1(config)# show bundle
- E. RP/0/RSP0/CPU0:PE1(admin)# show platform
- F. RP/0/RSP0/CPU0:PE1# show ipv4 interfaces brief

Answer: F

Explanation:

QUESTION NO: 73

Which Cisco IOS XR command is involved in enabling remote access?

- A. RP/0/RSP0/CPU0:PE1(config)# router static
- B. RP/0/RSP0/CPU0:PE1(config)# line console 0
- C. RP/0/RSP0/CPU0:PE1(config)# vrf default
- D. RP/0/RSP0/CPU0:PE1(config)# telnet ipv4 server max-servers 20
- E. RP/0/RSP0/CPU0:PE1(config)# username CISCO password LAB

Answer: D

Explanation:

QUESTION NO: 74

Which three physical lines can provide local or remote access to a Cisco router or switch?
(Choose three.)

- A. console port
- B. modem port
- C. virtual terminal lines
- D. switch fabric interface
- E. aux port
- F. managementEthernet0/RSP0/CPU0/0

Answer: A,E,F

Explanation:

QUESTION NO: 75

In Cisco IOS XR, which three operations is Admin mode used for? (Choose three.)

- A. Secure Domain Router management
- B. configuration-register settings
- C. hostname configuration
- D. IOS XR software package upgrades
- E. interface duplex settings
- F. domain-name configuration

Answer: A,B,D

Explanation:**QUESTION NO: 76**

Which two statements are correct regarding routing protocol authentication? (Choose two.)

- A. MD5 authentication can only be established between Internal BGP peers.
- B. Authentication can be established using keychains on one peer and MD5 hashing on the other.
- C. UDP port 179 is used for BGP session authentication.
- D. MD5 passwords are never sent to a peer.
- E. Routers authenticate each routing update packet received.
- F. BGP peers can be configured with different passwords.

Answer: D,E

Explanation:**QUESTION NO: 77**

Refer to the exhibit.

```
tacacs-server host 192.168.1.2
  key tacacskey
!
tacacs source-interface gigabitEthernet0/1/0/1
```

Which two statements are true regarding the following configuration?

- A. Users are authenticated using RADIUS.
- B. TACACS+ is configured for remote AAA.
- C. A TACACS+ AAA server group is configured.
- D. A TACACS+ server can be reached through the GigabitEthernet0/1/0/1 interface.
- E. Logging is configured to the local database.

Answer: B,D

Explanation: